

Abandoned Building Detection and Classification in Volos, Greece

Hon Kiu (James) TO

Department of Criminology, University of Pennsylvania

Feb 20, 2025

MUSA 6500 Deep Learning in Python

Design and implement an open-source software system for the detection of abandoned or disused buildings based on nighttime lights and built-up area indices via the combination of remote sensing data and machine learning algorithms.

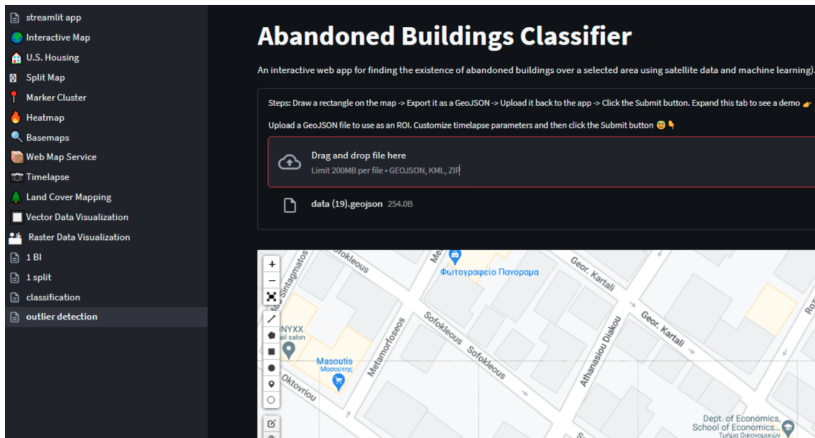


Figure 1: Snapshot of the abandoned building classifier user interface.

Impacts: Abandoned buildings and unmaintained structures are responsible for:

- ▶ Elevated crime rates (drugs, prostitution, etc.)
- ▶ Increased danger to public health and safety (prone to collapsing and fires due to deterioration, devaluation of nearby property values)
- ▶ Increasing costs for local government (security, inspection, additional police and fire services, etc.)
- ▶ Generating low property taxes

Overall, abandoned buildings contribute to a decline in a city's quality of life by providing an unappealing urban landscape for residents, visitors, and potential investors.

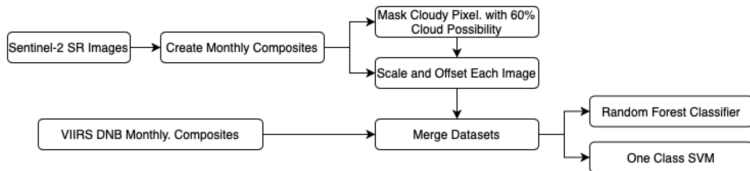


Figure 2: Machine learning pipeline diagram.

Model 1: Random forest (RF)

- ▶ Training dataset: Chicago, where ground-truth labels and exact geometry features of each building are available
- ▶ Requires labeled data for both abandoned and non-abandoned buildings

Model 2: One-class SVM (OSVM)

- ▶ Training dataset: Hand-labeled geo-referenced dataset for abandoned and non-abandoned buildings across the municipality of Volos
- ▶ Setting non-abandoned buildings as the positive class and assuming that buildings that did not fulfill all the criteria set by the OSVM as abandoned



Feature Selection

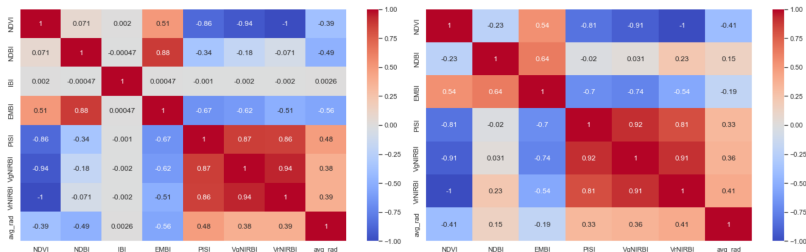


Figure 3: Correlation matrices for index time series generated for Chicago (left) and Volos (right).

Features	Parameters		RF Accuracy
	Max Depth	Number of Estimators	
NDVI, NDBI	20	400	66.3%
NDVI, NDBI, Average Radiance	20	350	68.1%
VgNIRBI, Average Radiance	20	600	66.6%

Table 1: Selected feature combinations for random forests.



Figure 4: Left: Experimentation results on all areas (25×25) using NDVI/NDBI (**top three**) and NDVI/NDBI/average radiance (**bottom three**) in random forests. Right: Experimentation results on all areas (25×25) using EMBI index (**top three**) and EMBI average radiance (**bottom three**) in One-Class SVM.

Discussions:

- ▶ **Top left:** Unable to not only detect abandoned buildings but also identify the non-abandoned buildings correctly
- ▶ **Bottom left:** Capable of detecting the majority of non-abandoned buildings but fail dramatically to identify the abandoned buildings
- ▶ **Top right:** Capable to detect some abandoned buildings but not to a satisfying degree
- ▶ **Bottom right:** The ability to detect abandoned and non-abandoned buildings seems to be the best among all considered models

Takeaways:

- ▶ Utilizes freely accessible satellite images and explore areas with scarce ground-truth data, allowing small and medium enterprises, and policymakers to leverage them
- ▶ Nighttime light intensity and built-up area indices are strong indicators, but models could be sensitive to nighttime light

Challenges:

- ▶ Limited computational power restricts very granular polygon-splitting
- ▶ Equal-sized polygons inhibit the ability to capture the characteristics of the area of interest
- ▶ Image resolution limits
- ▶ Hyperparameter tunings; currently is done manually instead of, e.g., grid search

Some questions for thought:

1. Can remote sensing be applied to assess the effectiveness of urban interventions (e.g., greening vacant lots, increased street lighting) in reducing crime?
2. Does the impact of nighttime lighting on crime differ across different urban settings (e.g., nightlife districts vs. residential areas)?
3. Could similar methods be adapted to detect emerging gentrification trends or areas vulnerable to economic decline?

Thank You!

Kyriakos, C., & Vavalis, M. (2023). Business Intelligence through Machine Learning from Satellite Remote Sensing Data. *Future Internet*, 15(11), 355